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Research on Multi-Sensor Data Fusion Technology of Artificial Intelligence in Robot Simulation

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***Abstract*—A multi-sensor information fusion method based on multi-sensor data is proposed. The motion control of the wheeled robotic arm is realized through the cooperation and complementation of multi-sensor information. The experimental results show that the designed driving device has good stability and can accurately control the wheel trajectory under certain test conditions. Attribute reduction, value reduction and other methods are used to remove redundant data in multi-dimensional data. The simplest decision criteria are extracted and a robot data fusion method based on multi-dimensional data is constructed. This can improve the intelligent perception capabilities of multiple robots.**

Keywords—Artificial intelligence, multi-sensor, data fusion technology, robot simulation application, rough set theory

I. INTRODUCTION

In recent years, with the continuous development of science and technology, perception and control system has gradually become a hot topic in the world, and has received the attention of many research institutions and enterprises, and its research results and its future development have played a crucial role. In short, the robot has experienced three stages of teaching and representation, sensory and intelligent, from having no perception of the external environment, to having sensory abilities such as touch, vision and hearing, and finally to being able to communicate with the outside world, cooperate and autonomously judge and execute [1]. In order to make the robot intelligent, the sensors corresponding to the five senses and operation capabilities of people, such as touch, vision, position, vibration, infrared, etc., are positioned and manipulated to achieve similar feelings and actions with people. In recent years, in industrial production, a variety of sensors are increasingly widely used, however, due to different application occasions and different detection distances, so that the information they provide is limited to a specific environment or object, therefore, the detected information must be integrated analysis in order to better play its role. Information fusion between multiple sensors, also known as data fusion, refers to the automatic analysis and synthesis of the information and data detected by multiple sensors according to specific standards through computer technology, so as to achieve the judgment and judgment of the target.

II. REVIEW OF MULTI-SENSOR DATA FUSION METHODS

It detects a variety of specific information, when it detects some special information, it will be in accordance with the internal rules of some information, the perceived information into a certain format, and then output the information obtained, whether it is an electrical signal or the required signal of the device where the sensor is. Therefore, the use of sensors for information collection, processing, storage, transmission, display and control can better achieve the transmission of information [2]. With the development of modern science and technology, various scientific and technological methods emerge endlessly, and various sensors are also updated. A single sensor can only detect a single piece of information, and can only carry out simple control of intelligent products, and cannot achieve intelligence. Therefore, in the field of artificial intelligence, a variety of different sensors are often used, and different information is integrated through different sensors to build a neural network with autonomous cognitive function. Therefore, it is necessary to use the data of multiple sensors to fuse and form intelligent products with intelligent characteristics. Multi-sensor information fusion refers to the integration of various information and data from various sensors through computer technology, and the data algorithm under big data and cloud computing can automate the analysis and fusion of various information, so as to make the right choice for intelligent products in the current information environment. In fact, multi-sensor information fusion is a method that organically combines information fusion and neural network. It is like people can make independent judgments based on the obtained information after obtaining various kinds of information.

III. ROBOT SIMULATION EXPERIMENT SYSTEM BASED ON MULTI-SENSING TECHNOLOGY

A. Selecting the Information Fusion Architecture Mode

In view of the problems existing in multi-source sensor networks, the sensor fault diagnosis problem in sensor networks is studied. The model can be divided into functional model, structural model and mathematical model [3]. Function module mainly introduces the main functions of the system in detail, and each module in the system is introduced in detail. The whole model consists of two stages: fusion algorithm and logic operation. According to the degree of data decomposition, it can be divided into: feature layer, data layer and decision layer. (Figure 1 is quoted in Developing a Reliability Model of CNC System under Limited Sample Data Based on Multisource Information Fusion).

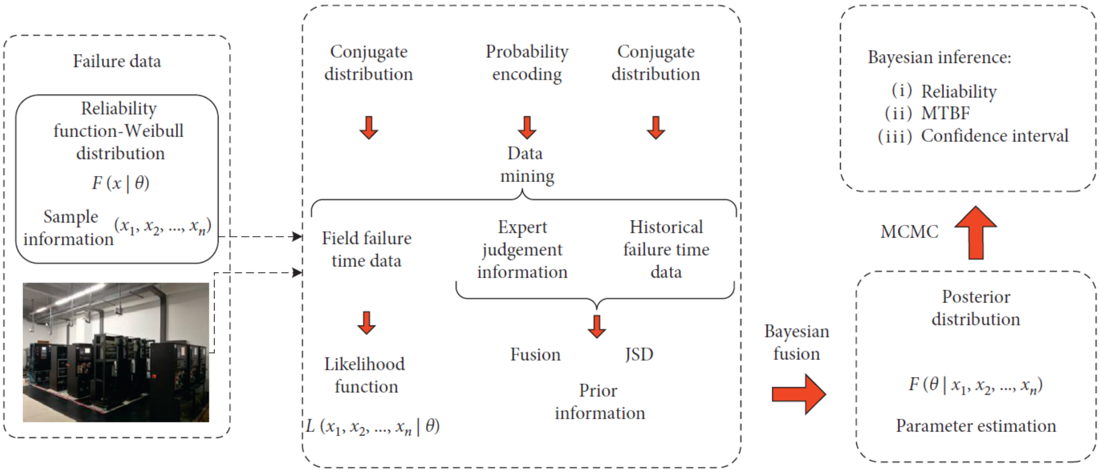

Fig. 1. Structural model of information fusion

The fusion algorithm based on the feature layer is selected to achieve the target, and the index of information loss, fault tolerance, computation, fusion degree, real-time, accuracy and anti-interference reaches the medium level [4]. After the main control unit of the robot arm collects the information of each sensor, it adopts the fusion mode of feature layer and selects the appropriate data fusion algorithm to process and identify the collected information after various information fusion processing, and outputs it to the control system of the robot arm to achieve the best control effect.

B. Methods for simulating mechanical arm with Rough Set

In multi-sensor information fusion, there are some problems such as complementarity, contradiction, fuzziness and certainty among different sensors. Therefore, how to analyze and optimize the multi-path and multi-space-time perception information and comprehensively reduce its inconsistency, depict the target characteristics more comprehensively, and improve the rate and accuracy of information fusion is a difficult problem that needs to be studied and solved [5]. Through the processing of incomplete, inconsistent and inaccurate data, the essential relationship is mined and effective features are extracted to realize the processing of data. Therefore, a highly simplified decision criterion can be extracted by using rough set method for multi-sensor information fusion, and an efficient and fast information fusion algorithm can be obtained.

(1) The information detected by the corresponding sensor is pre-processed into a unified data storage pattern. (2) Construct an information system S= (U, A, V, f), where U= {x1, x2, x3... xn} is the amount of information obtained by the sensor, determines the state and determination characteristics of the information set, and constructs the information table. (3) Obtain the list of core values of the model. (4) A simplified information table is obtained from the list of kernel values, so as to obtain a decision table. (5) When the state characteristic in the table is judged to be continuity, it is discretized. (6) The concept of attribute reduction and kernel is used to reduce the attributes of the decision table, remove the redundant state attributes and repeated attributes, and obtain the reduced decision table. (7) Determine the extraction of decision rules. (8) Summarize the decision rules extracted in step (7), the corresponding multi-sensor information fusion algorithm. The algorithm flow is shown in Figure 2 (Picture quoted from Special Issue "AI Algorithm Design and Application").

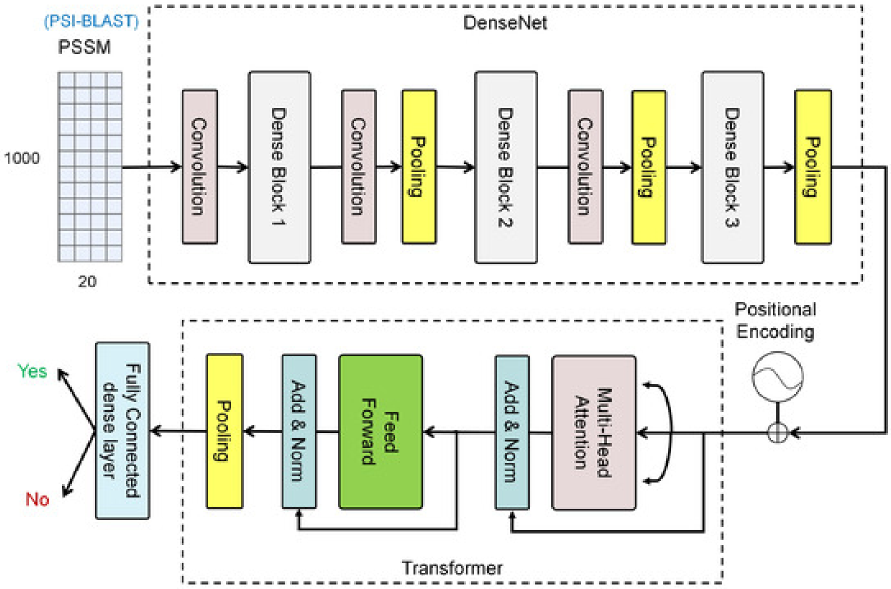

Fig. 2. Algorithm flow based on attribute reduction

C. System Process

Mobile robot MR Is a kind of movable robot with specific intelligence. In order to be able to sense the surrounding situation, so as to achieve the purpose of obstacle avoidance, it is necessary to use sensors to detect and collect information about the surrounding environment. Due to the complexity of the environment, it is difficult for a single sensor to provide sufficient, accurate and real-time environmental information [6]. Use multiple sensors for more accurate detection of the surrounding situation and improve the cognitive ability of the surrounding environment. However, there are a lot of redundant and conflicting information in multiple sensors. Multi-sensor data fusion is a better method. The multi-sensor data fusion method introduced above is applied to this problem. The above method is tested by simulation experiment [7]. With the square black dot as the obstacle, the hollow circle representing the robot, and the solid line being the target to be reached. Setting up a virtual environment is shown in Figure 2. Suppose that from the beginning A to the end B, the process encountered a variety of obstacles, in order to reach the end B, you must avoid the obstacles along the way. As can be seen from FIG. 2, the robot can smoothly bypass the obstacle to reach the end B (the picture is quoted in Energies 2021, 14(2), 327).

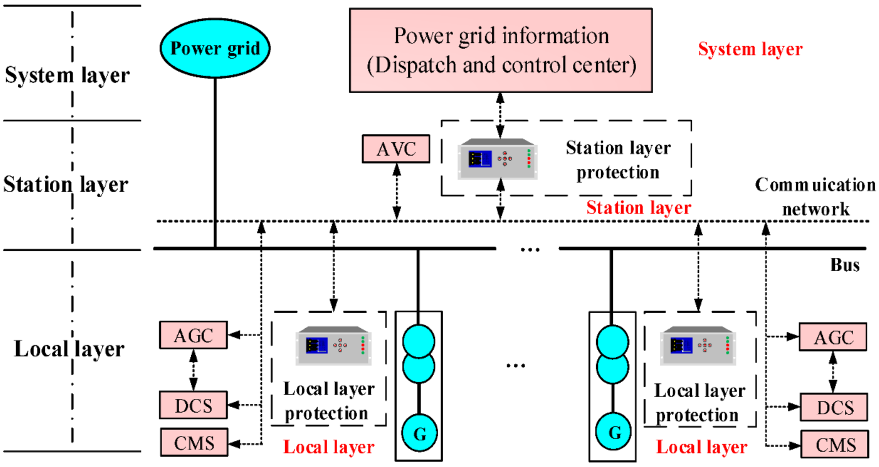

Fig. 3. Flow of robot obstacle avoidance algorithm based on rough set

Since the mathematical model of the robot is nonlinear, and there are many input and output variables, and there are coupling between the variables, its control process includes multiple levels such as task planning, trajectory planning, servo control and action planning [8]. Its control sequence flow is shown in Figure 3-4 (the picture is quoted in Advanced Engineering Informatics, Volume 51, January 2022, 101507). The robot first obtains the operator's instructions through the man-machine interface, which can be in the form of input signals such as touch screen, mobile phone or keyboard, or commands issued by human natural language. The robot interprets and interprets the input control commands, decomposes the operator's commands into machine language, and completes task planning. In order to realize the autonomous intelligent walking of the robot, the motion route of the robot is designed and the path planning is completed [9]. Obstacles encountered in the process of movement can be avoided to achieve robot obstacle avoidance; For a robot with a robot arm, various movements of the robot arm and the robot body are planned according to the needs.

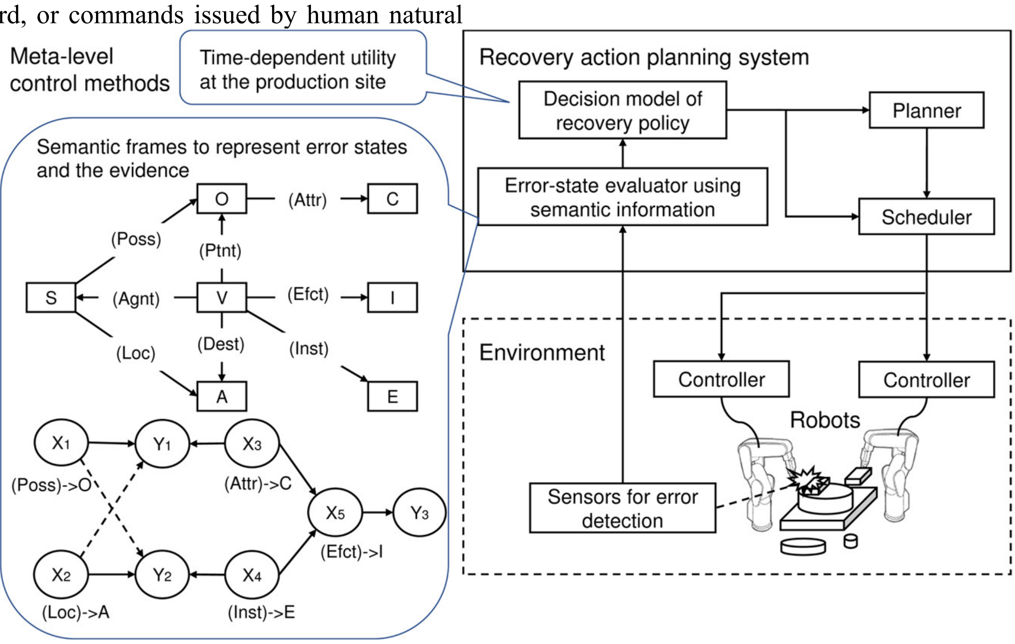

Fig. 4. Sequence flow of robot control

IV. System Inspection

In the real environment, the angles of the six light mice are all 60 degrees, so the direct results are not comparable, so we preprocessed the position information of each light mouse in cartesian coordinates. By conducting experiments on 10 representative samples, 10 samples of different types were obtained, and the results are shown in Table I and Figure 5.

TABLE I. Effect of light mouse test and information fusion under 6 different preconditions

10 sets of original measurements	**1**	**2**	**3**	**4**	**5**
Photoelectric mouse sensor 1	108.8	86.7	50.5	59.1	22.1
Photoelectric mouse sensor 2	108	86.4	51.3	27.3	22.8
Photoelectric mouse sensor 3	107.4	24.6	49.9	59	96
Photoelectric mouse sensor 4	52.9	86.5	50.8	59.6	22.3
Photoelectric mouse sensor 5	108.4	85.8	50.3	59.2	21.4
Photoelectric mouse sensor 6	107.8	86.1	133	58.8	21.6
Data fusion result	106.8	84.6	52.5	58.4	24.3
Six sensor averages	98.9	76	64.3	53.9	34.4
Actual truth value	108.1	86.4	50.5	59.1	22
10 sets of original measurements	6	7	8	9	10
Photoelectric mouse sensor 1	49.3	79.3	41	47.2	18.9
Photoelectric mouse sensor 2	100.9	78.9	41.4	68.1	18.3
Photoelectric mouse sensor 3	102.1	78.6	82.5	67.6	20
Photoelectric mouse sensor 4	101.7	79.5	40.2	67.4	73.8
Photoelectric mouse sensor 5	102.5	37.4	41.9	67.8	21.1
Photoelectric mouse sensor 6	101.9	78.6	40.5	68.3	21.6
Data fusion result	100.4	78	42	67.4	21.4
Six sensor averages	93	72.1	47.9	64.4	29
Actual truth value	101.8	79	41	67.8	20

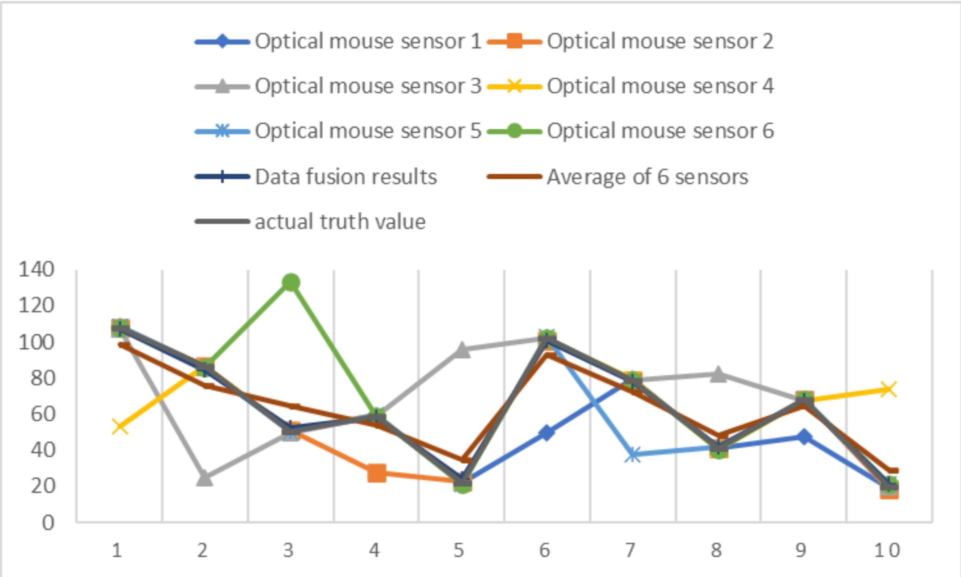

Fig. 5. Simulation results

The average value of the measured data of 6 light mice is analyzed, and the fusion results of 10 groups of target truth values, arithmetic mean values are obtained. The method obtained by this method is very close to the simple arithmetic mean value when each light mouse is accurately detected. However, if there is a large error or some deviation in the detection of some light mice, the above three lines are analyzed, and it can be seen that the information obtained by the multi-sensor information fusion method in this paper can better achieve the real purpose than the simple arithmetic average, thus greatly reducing the interference caused by this situation. The usual data processing method is that, when the arithmetic average of the measurement results, the data with a large deviation must be removed first, that is, a threshold is set to remove the data that is too different from the data measured by other data sensors. The detection accuracy of the ranging method based on this technology mainly depends on its own moving condition. Therefore, when the action state of the mechanical arm changes dramatically, it is impossible to remove it by setting the threshold value. In this case, the proposed multi-sensor information fusion method can obtain high detection accuracy even if all abnormal information is not removed.

V. Conclusion

In this paper, the motion control technology of wheeled robots is studied based on the multi-sensor information fusion model. Due to the complexity of the environment and the defects of robot perception, it is difficult for robots to complete the set tasks when using a single sensor. However, blindly stacking sensors cannot essentially improve the system performance, but will increase the burden of the system. This project intends to adopt the. After program debugging and verification, this method can realize arbitrary movement within 40 cm-100 cm, and the trajectory radius can be flexibly adjusted. At about 5cm away from the stage, the system can detect the edge of the stage and then back or turn accordingly according to its own posture. Under certain experimental conditions, the motion control of the robot has good stability.

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REFERENCES

[1] Tian Yajuan, Fu Daping, Wu Shihui. Research on the application of multi-sensor information fusion technology in robots. Automation and Instrumentation, vol. 38, pp.51-53, February 2022.

[2] Li Weidong, Xie Zaijin, Li Min. Research on robot sweeping algorithm based on multi-sensor fusion. Electronic Design Engineering, vol. 29, pp.61-69,February 2022.

[3] Zhang Weixing, Dang Chaoqun, Zhang Suoping, et al. Research on wave spectrum data acquisition and transmission based on multi-sensor fusion. Ocean and Limnology, vol. 54, pp.645-652,March 2022.

[4] Li Lirui, Zhang Jiacheng, Zhang Borui. Research on knowledge service of integrated intelligent robots from the perspective of scientific intelligence *. Books and Information, vol. 43, pp.61-68,February 2022.

[5] Zhang Huaijie, Ma Jingya, Liu Haoyuan, et al. Research on indoor autonomous positioning technology of multi-rotor flying robot based on vision and inertia fusion. Acta Aeronautica et Astronautica Sinica, vol. 44, pp.134-139,May 2021.

[6] Yuan Chao, Wang Shaojun, Jiang Yang, Wang Zhu, Zhai Yongjie. Design and simulation verification of data glove for robot teleoperation. Instrument Technology and Sensors, vol. 11, pp.47-56,September 2022.

[7] Leng Fang, Pei Zhouqi. Navigation system of underwater vehicle based on multi-sensor information acquisition. Ship Science and Technology, vol. 45, pp.97-100, August 2022.

[8] Zhang Shuliang, Tan Xiangquan, Wu Qingwen. Research on indoor mobile robot localization based on multi-sensor fusion technology. Sensors and Microsystems, vol. 40, pp.44-48,August 2022.

[9] Chang Li, Chang Miao. Navigation system design of submarine mapping robot based on multi-sensor information fusion. Automation and Instrumentation, vol. 38(3):35-39,March 2021.